Emma Huang

B.S. Electrical Engineering - ML and Controls

510-960-9858 | emmaymhuang@gmail.com | Google Scholars | LinkedIn | GitHub | Website

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EDUCATION

University of California, San Diego 2023–2027
B.S. Electrical Engineering *San Diego, CA*

- **Degree Specialization:** Machine Learning and Controls (GPA: 3.961)
- **Honors:** Provost Honors (all quarters with 12+ graded units); Philip and Elizabeth Hiestand Scholarship

PUBLICATIONS

/ denote equal contribution*

- **Emma Huang**, Z.-Y. Chiu, N. Joglekar, S. Liu, and M. C. Yip, "Reeling it in: Flexible needle pick up via thread manipulation for autonomous suturing," *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2026. [Online]. Available: https://arxiv.org/abs/2607.26337
- **Emma Huang**, O. Hernandez, Z.-Y. Chiu, and M. Yip, "Live thread reconstruction for autonomous manipulation," *IROS DOM-R3 Workshop*, 2026. [Online]. Available: https://arxiv.org/abs/2610.30694
- **Emma Huang**, O. Hernandez, Z. Liang, Z.-Y. Chiu, and M. Yip, "Real-time thread reconstruction for autonomous suture thread and needle manipulation," *In review*, 2026.
- **Emma Huang***, M. Bohley*, K. Jamakhandi**, R. Fellbaum**, S. Tang, S. Tutton, A. Hsiao, A. Norbash, C. Chen, D. Chen, A. Rao, L. Peiros, and M. C. Yip, "Integrated respiratory sensing and gated control for autonomous needle insertion via ct bore robotic arm," *In review*, 2026. [Online]. Available: https://arxiv.org/abs/2610.30452

PRESENTATIONS

Undergraduate Research Conference Apr 2026
La Jolla, CA

- Presented: *What is an effective method to autonomously pick up suture needles?*

Summer Research Conference (SRC) Aug 2026
La Jolla, CA

- Presented: *How can robots suture autonomously?*

IEEE/RSJ IROS Sept 2026
Pittsburgh, PA

- Paper: *Reeling It In: Flexible Needle Pick Up via Thread Manipulation for Autonomous Suturing*

ACADEMIC EXPERIENCE

Advanced Robotics Controls Lab Mar 2024–Present
Undergraduate Researcher *La Jolla, CA*
Mentors: Zih-Yun Chiu, Michael Yip

- **Image to robot motion pipeline for suture needle pick up utilizing the thread.** Utilized segmented and reconstructed tissue point cloud, and spline representation of suture thread to design a contact aware grasp point optimizer, and perform a sequence of motions to lift and grasp needle. Experiments performed on the da Vinci Research Kit (dVRK); 61% full pipeline success, 91% reconstruction success. ***Reeling It In: Flexible Needle Pick Up via Thread Manipulation for Autonomous Suturing****, accepted to IROS 2026.*
- **Real Time Thread Reconstruction and Surgical Manipulation Tasks.** Build upon previous methods for autonomous thread reconstruction with real-time tracking and manipulation for needle pick up, thread detangling, and tightening tasks. **25.8% improvement** in needle pick up trials, and average **88.9% success** on task completion. Mentored an undergraduate student working on image segmentation. *Under review for RAL journal.*
- **Autonomous Needle Insertion for Lung Tumor Biopsy:** Led a **team of 12** undergraduates, Masters, and PhD students; Designed and built a torso phantom and needle insertion **testbed** and conducted live breathe tracking and gated needle insertion sequences. Resulted in 29.8% reduction in breath tracking error and successful needle insertions across all trials. *Under review for ICRA 2027.*

SEDS UCSD (Lander Project) Sep 2023–Jun 2026
Director of Electrical (2025), Avionics Lead (2024), Avionics Member (2023) *San Diego, CA*

- Designed and built active **buck boost converter** (30A/24V) for high power gimbal and valves power on **Lander** - bi-propellant LOX-LNG sounding vehicle.
- Avionics system integration: engine control unit, power control unit, flight computer; Designed custom PCB boards via Altium, EAGLE circuit design and Fusion 360 circuit housing.
- Outreached to local communities through tabling events - Air & Space Day, school trips, workspace tours.

Motion Predictive Controls Lab - UC Berkeley Jun 2023–Sep2023
Research Assistant *Berkeley, CA*

Mentor: Tony Zheng

- Built on pre-existing robot infrastructure to develop new algorithms for image processing and implementation on robot arms.
- Implemented motion capture controls on Fanuc and Universal robotic manipulators with Ridgeback and Turtlebot drive bases; tested motions using Gazebo, Rviz, robosuite, and universal robots simulations; wrote python and C++ code for robotic manipulator sketching on tabletop and canvas; fixed and improved robotic manipulator visualization and simulation.

Lawrence Berkeley National Laboratory Jun 2022–Aug 2022
Machine Learning Intern *Berkeley, CA*

Advisor: Doru Thom Popovici

- Researched the effect of different computer chips and network architecture in convolutional neural networks (CNN) using TensorFlow and PyTorch frameworks on the CIFAR-10 dataset.
- Results showed TensorFlow framework with a higher accuracy, while PyTorch had up to 4 times faster processing speeds.

PROFESSIONAL EXPERIENCE

ExtremeWaves May 2025–Jul 2025
Electrical Engineering Intern (RF Power PCB Design) *San Diego, CA*

- Designed, manufactured and tested high performance, 800 MHz **microcontroller** life support PCBs with Cadence Allegro; multi channel **SPI** interface; multi-power supply time sensitive boot up and shut down sequence; high frequency impedance control; quick verification and iterative building.
- Sketched three RF, differential signal six layer **transmission boards** with tight impedance and length matching.

Fabric8labs Jun 2024–Aug 2024
Electrical Engineering Intern (Current Sensor Design – PCB Testing) *La Jolla, CA*

- Debugged and repaired 30+ copper metal 3D printer controller PCBs comparable to **$15,000+**; troubleshooting and root cause analysis on signals (I2C, CAN, UART), power supplies, relays, sensors, and FPGA (Kria) components.
- Improved JupyterLab printer Python **testing scripts** used to recover printer controller PCBs with **automated** board failure detection; Developed user interface for the testing scripts with IPyWidgets.
- Designed, tested, and implemented **current and voltage sensors** with **Altium** into fleet of 40+ printer controller PCBs with PICO and C++; Software debugging and repository update for the printhead controllers.

Scripps Sandbox Makerspace Mar 2024–Mar 2025
Makerspace Technician *La Jolla, CA*

- Improve and maintain makerspace infrastructure including the setup of 3D-printers, custom shelving, safety infrastructures, etc.
- Technical consultation for Scripps faculty and UCSD affiliates; Organize collaborations with Scripps faculty, campus organizations, student clubs, and more.
- Developed, organized, and led 15 person mood lamp workshop teaching CAD, soldering, and 3D-printing; 8 person plastic leather workshop teaching heat press, recycling, leather crafting, and design; Assisted workshops teaching laser cutting, drilling, and Inkscape (2D drawing).

University of California - Berkeley (ADTP) Jul 2023
Substitute Teacher *Berkeley, CA*

- Advanced Robotics Engineering; two sessions; 3.5 hr each; 24 students; 7/8/9th grade.
- Assisted students' final projects including robotics arm gripper, self balancing robot, and line tracing race car.
- Kept students engaged; answered technical questions; communicated logistics with teacher and staff.

ECE 15 - Intro to C Programming Jan 2025–Mar 2025
Tutor and Reader *La Jolla, CA*

- 10 hours of tutoring sessions per week.
- Guide students on introductory C programming coding assignments.

VOLUNTEER EXPERIENCE

Envision Makerspace, UC San Diego Oct 2023–Apr 2024
Makerspace Staff *San Diego, CA*

- Assist students with soldering, 3D-printing, laser cutting, wiring, debugging, and shop tools.
- Maintenance of shop equipment.

SEDSxFEMxTR Aug 2024–Jun 2025
Mentee 2023, Mentor 2024 *San Diego, CA*

- Provide technical, educational, and community mentorship to first year female engineers to increase retention rate of engineers of diverse gender backgrounds.
- Conducted Weekly community discussions, one-on-one meetings, and supported professional events.